

Fundamental Math/Physics for Computer Graphics (II)

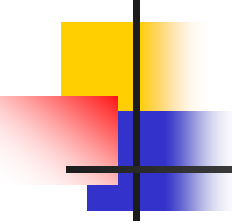
Lecture 4

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DEPARTMENT OF COMPUTER SCIENCE



Deformations

Transformations that do not preserve shape

- Non-uniform scaling
- Shearing
- Tapering
- Twisting
- Bending



Shearing

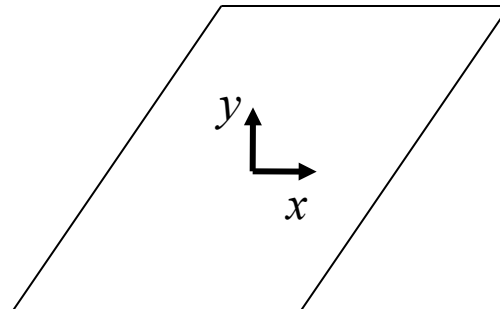
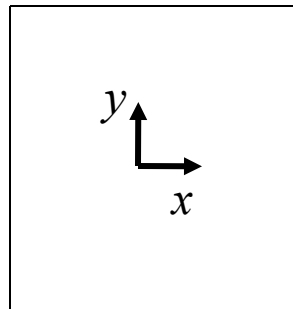
$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & s_{xy} & s_{xz} & 0 \\ s_{yx} & 1 & s_{yz} & 0 \\ s_{zx} & s_{zy} & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

$$s_{xy} = 1$$

$$s_{xz} = 0$$

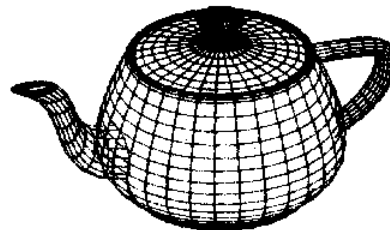
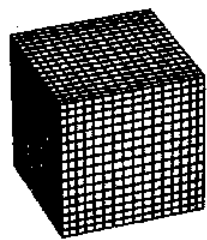
$$s_{yx} = s_{yz} = 0$$

$$s_{zx} = s_{zy} = 0$$

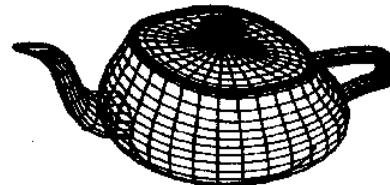
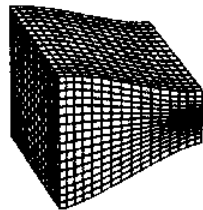


Tapering

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & f(x) & 0 & 0 \\ 0 & 0 & f(x) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$



Original objects

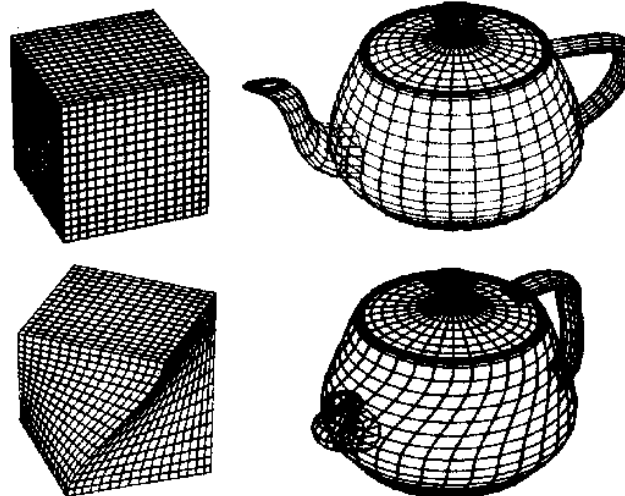


Tapering



Twisting

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos(\theta(y)) & 0 & \sin(\theta(y)) & 0 \\ 0 & 1 & 0 & 0 \\ -\sin(\theta(y)) & 0 & \cos(\theta(y)) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$



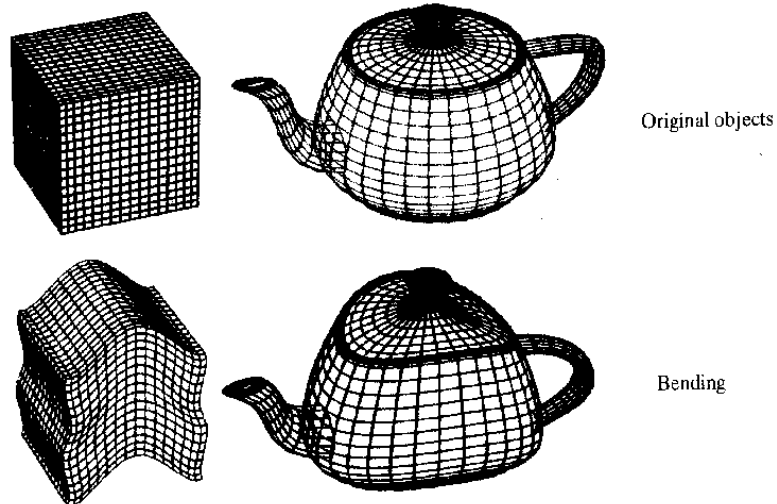
Original objects

Twisting

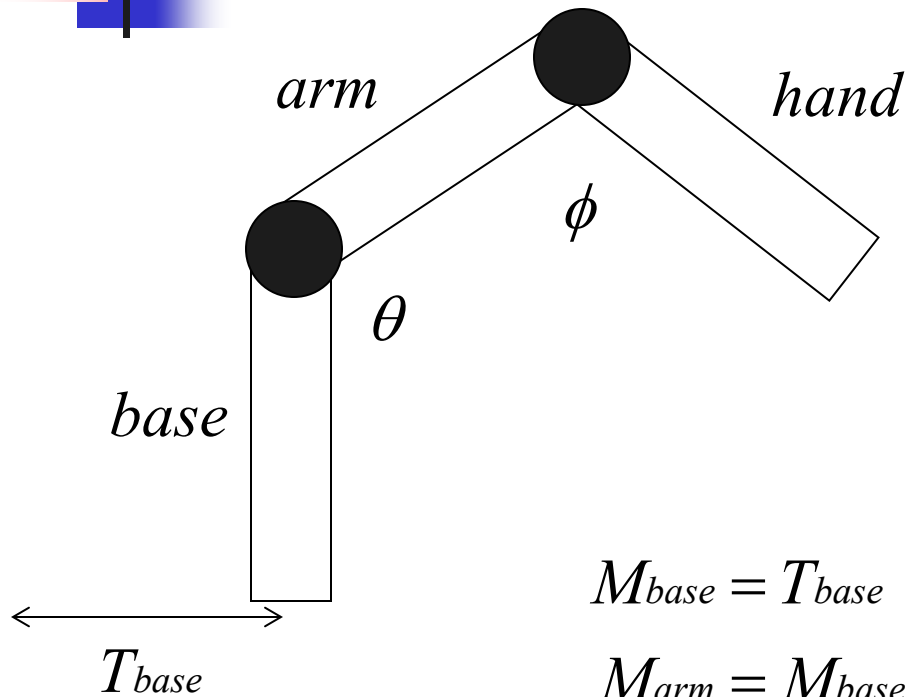


Bending

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & f(y) & g(y) & 0 \\ 0 & h(y) & k(y) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$



Hierarchical Transformations



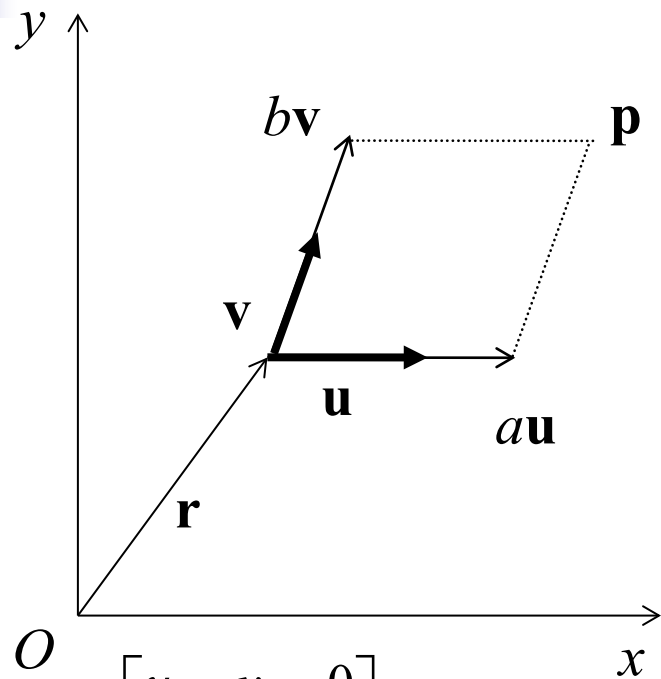
$$M_{base} = T_{base}$$

$$M_{arm} = M_{base} R(\theta) T_{arm} = T_{base} R(\theta) T_{arm}$$

$$M_{hand} = M_{arm} R(\phi) = T_{base} R(\theta) T_{arm} R(\phi)$$



Coordinate Systems



$$M = \begin{bmatrix} u_x & v_x & 0 \\ u_y & v_y & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\text{and } T = \begin{bmatrix} 1 & 0 & r_x \\ 0 & 1 & r_y \\ 0 & 0 & 1 \end{bmatrix}$$

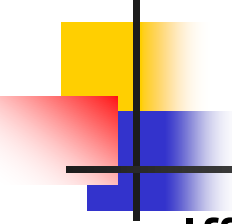
$$\mathbf{p} = \mathbf{r} + a\mathbf{u} + b\mathbf{v}$$

$$\begin{bmatrix} p_x \\ p_y \\ 1 \end{bmatrix} = T M \begin{bmatrix} a \\ b \\ 1 \end{bmatrix}$$

$$T^{-1} \begin{bmatrix} p_x \\ p_y \\ 1 \end{bmatrix} = M \begin{bmatrix} a \\ b \\ 1 \end{bmatrix}$$

$$M^{-1}T^{-1} \begin{bmatrix} p_x \\ p_y \\ 1 \end{bmatrix} = \begin{bmatrix} a \\ b \\ 1 \end{bmatrix}$$





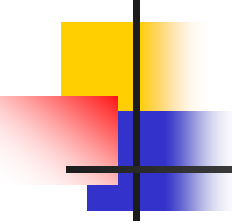
Orthonormal Coordinates

Iff $\mathbf{u}$ and $\mathbf{v}$ are orthonormal:

$$M^{-1} = \begin{bmatrix} u_x & u_y & 0 \\ v_x & v_y & 0 \\ 0 & 0 & 1 \end{bmatrix} = M^T$$

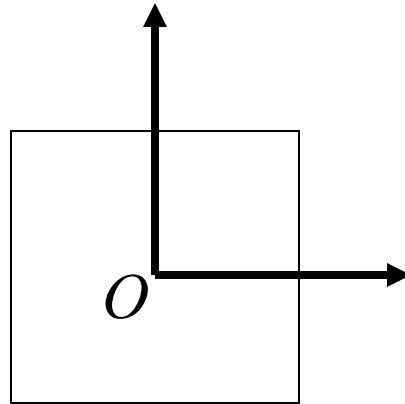
$$M^{-1}T^{-1} = \begin{bmatrix} u_x & u_y & 0 \\ v_x & v_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -r_x \\ 0 & 1 & -r_y \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} u_x & u_y & -\mathbf{r} \cdot \mathbf{u} \\ v_x & v_y & -\mathbf{r} \cdot \mathbf{v} \\ 0 & 0 & 1 \end{bmatrix}$$

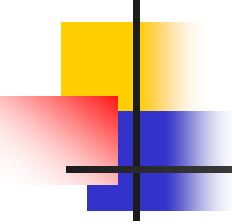




Object Coordinates

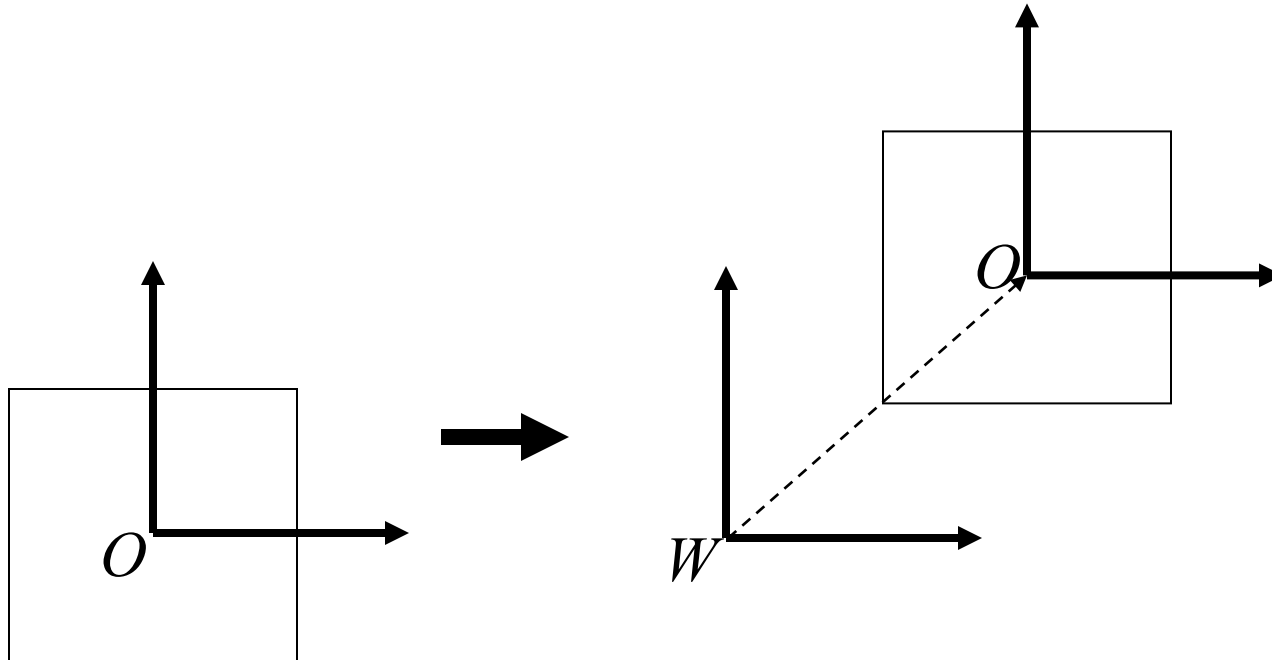
Convenient place to model the object

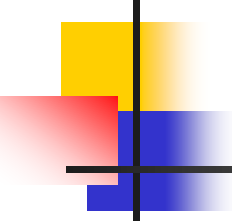




World Coordinates

Common coordinates for the scene

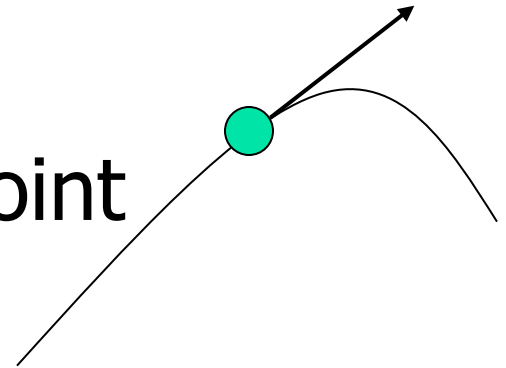




Dynamics

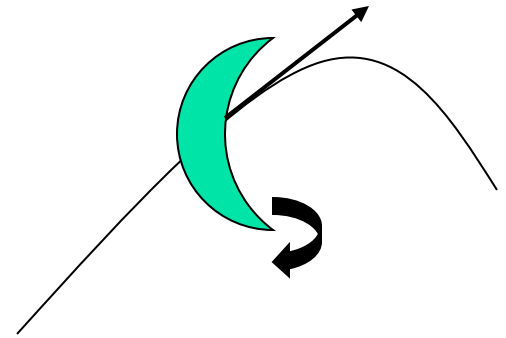
- Dynamics of a material point

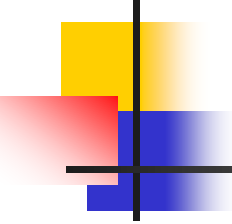
Ideal case, but often sufficient



- Dynamics of a solid

Including rotation, torques...





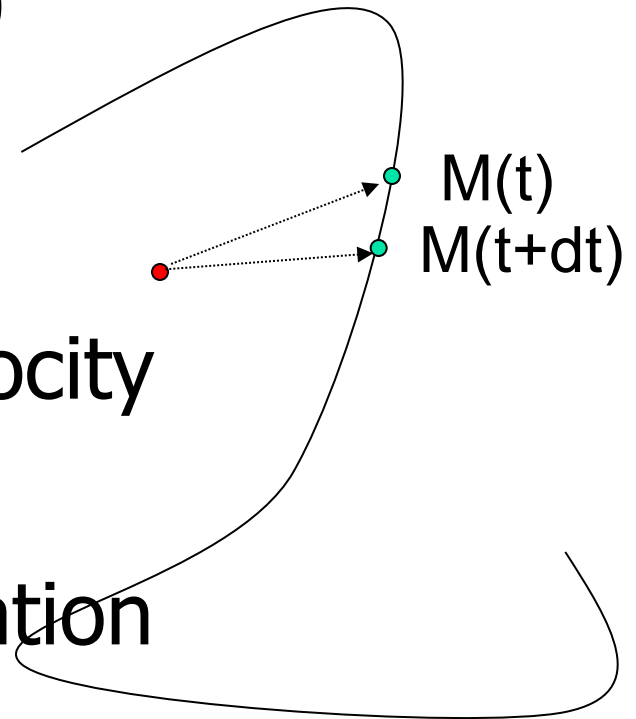
Position, Velocity, Acceleration

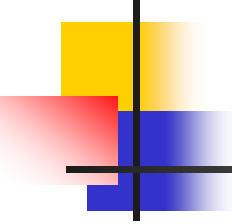
$$\mathbf{V} = \lim_{dt \rightarrow 0} \frac{\mathbf{OM}(t+dt) - \mathbf{OM}(t)}{dt}$$

$$\mathbf{V} = d\mathbf{OM}/dt = \dot{\mathbf{OM}} = \text{velocity}$$

$$||\mathbf{V}|| = \text{speed}$$

$$\begin{aligned}\mathbf{A} &= d\dot{\mathbf{V}}/dt = \dot{\mathbf{V}} = \text{acceleration} \\ &= d^2\mathbf{OM}/dt^2 = \ddot{\mathbf{OM}}\end{aligned}$$





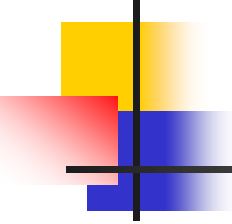
Newton Law (*material point*)

$$\mathbf{F} = m \mathbf{A}$$

F: sum of the forces exerted

m: mass of the material point



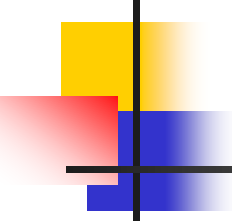


Exercise: Material point falling

- $\mathbf{A}(t) = \mathbf{g}$
- $\mathbf{V} = \mathbf{V}_0 + \mathbf{g} t$
- $\mathbf{M} = \mathbf{M}_0 + \mathbf{V}_0 t + .5 * \mathbf{g} t^2$

Now, how can we find the motion of this point when some external forces are present?





Temporal integration

- In the previous example, we were able to explicitly integrate the motion.
- However, we must often numerically integrate the motion – For instance, Newton method:

$$\mathbf{A}(t) = \mathbf{F}(t) / m$$

$$\mathbf{V}(t) = \mathbf{V}(t-dt) + \mathbf{A}(t) dt$$

$$\mathbf{M}(t) = \mathbf{M}(t-dt) + \mathbf{V}(t) dt$$

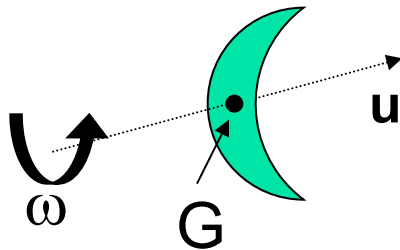


Point vs. Object

- We overlooked rotations! A real object can also spin on itself during motion.
- Notion of **angular velocity**:

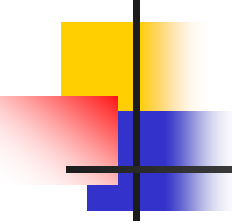
$$\boldsymbol{\omega} = \omega \mathbf{u}$$

angular speed axis of rotation



(G: center of mass)





Integration

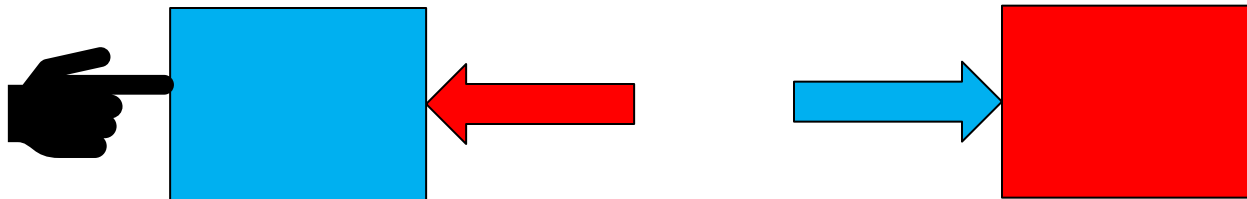
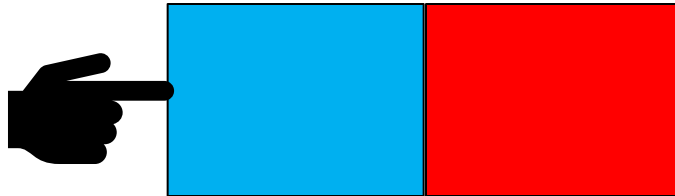
- Just a bit more complex...
- Quaternion make it easier
- We will review it later (rigid body simulation)

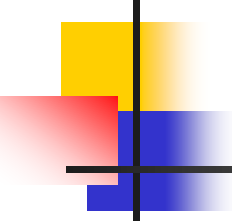


Other things to know

- Action/Reaction Principle:

$$\mathbf{F}_{1/2} = - \mathbf{F}_{2/1}$$





2D Triangles

- Definitions: given 2D points a, b, c
- Area:
 - Positive
 - Negative (clockwise)

$$\begin{aligned}\text{area} &= \frac{1}{2} \begin{vmatrix} x_b - x_a & x_c - x_a \\ y_b - y_a & y_c - y_a \end{vmatrix} \\ &= \frac{1}{2} (x_a y_b + x_b y_c + x_c y_a - x_a y_c - x_b y_a - x_c y_b) .\end{aligned}$$



Barycentric Coordinate

$$\mathbf{p} = (1 - \beta - \gamma)\mathbf{a} + \beta\mathbf{b} + \gamma\mathbf{c}.$$

$$\alpha \equiv 1 - \beta - \gamma,$$

$$\mathbf{p}(\alpha, \beta, \gamma) = \alpha\mathbf{a} + \beta\mathbf{b} + \gamma\mathbf{c},$$

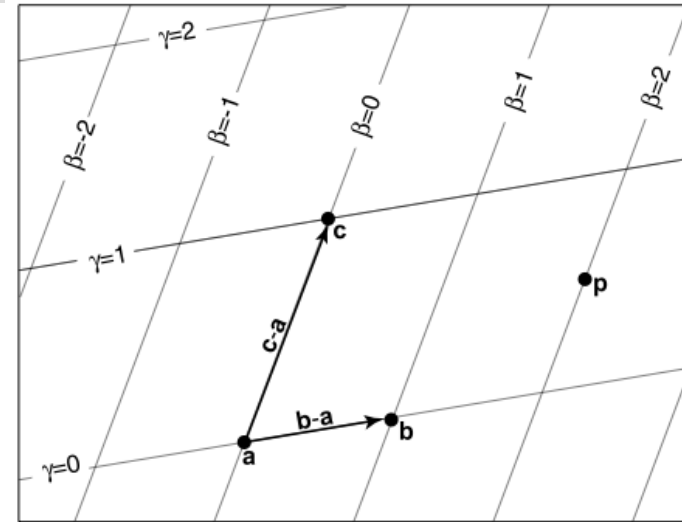
$$\alpha + \beta + \gamma = 1.$$

A point is inside the triangle if and only if

$$0 < \alpha < 1,$$

$$0 < \beta < 1,$$

$$0 < \gamma < 1.$$



$$\mathbf{p} = \mathbf{a} + \beta(\mathbf{b} - \mathbf{a}) + \gamma(\mathbf{c} - \mathbf{a}).$$



Barycentric Coordinate II

$$\gamma = \frac{(y_a - y_b)x + (x_b - x_a)y + x_a y_b - x_b y_a}{(y_a - y_b)x_c + (x_b - x_a)y_c + x_a y_b - x_b y_a}.$$

$$\beta = \frac{(y_a - y_c)x + (x_c - x_a)y + x_a y_c - x_c y_a}{(y_a - y_c)x_b + (x_c - x_a)y_b + x_a y_c - x_c y_a},$$

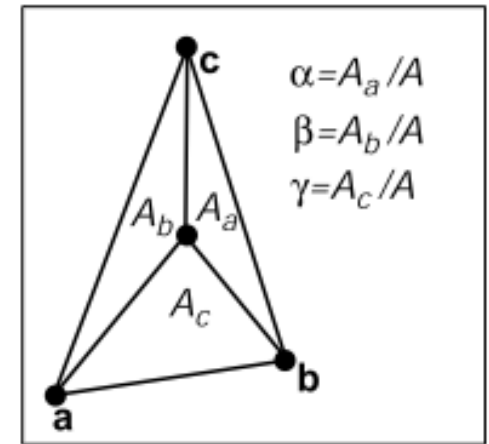
$$\alpha = 1 - \beta - \gamma.$$

Another way to compute Barycentric coordinates

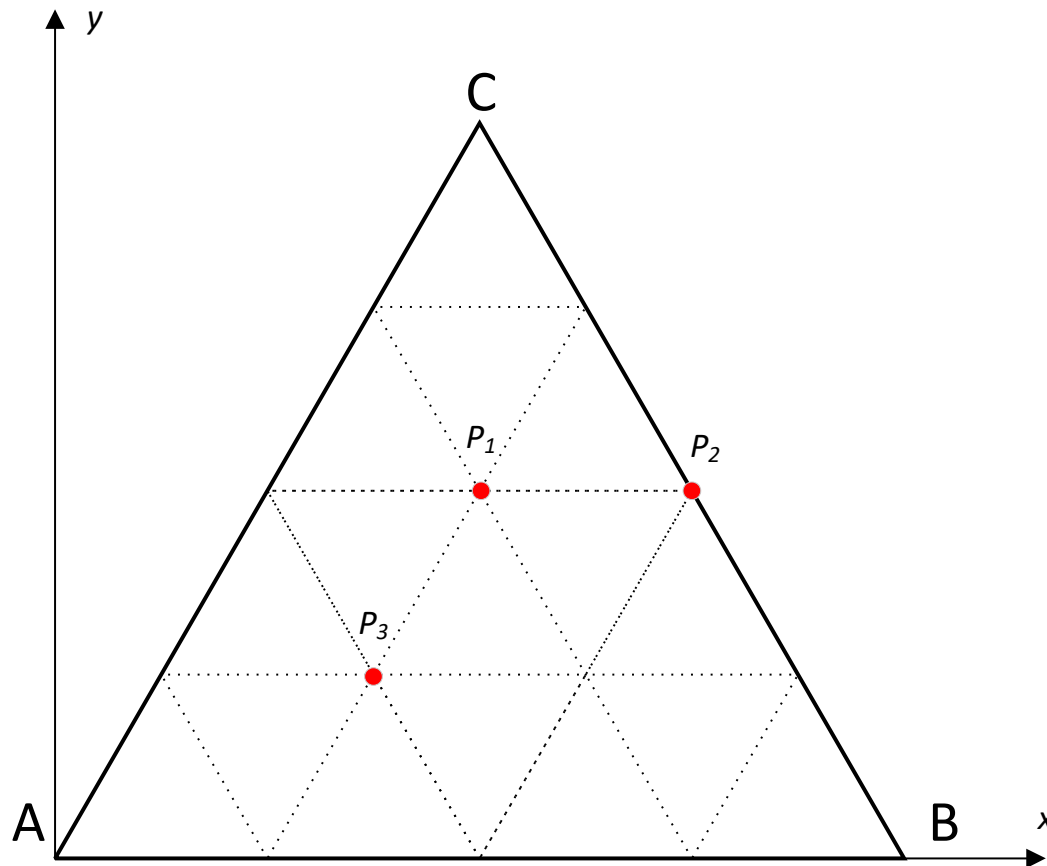
$$\alpha = A_a/A,$$

$$\beta = A_b/A,$$

$$\gamma = A_c/A,$$



Coordinate Example I

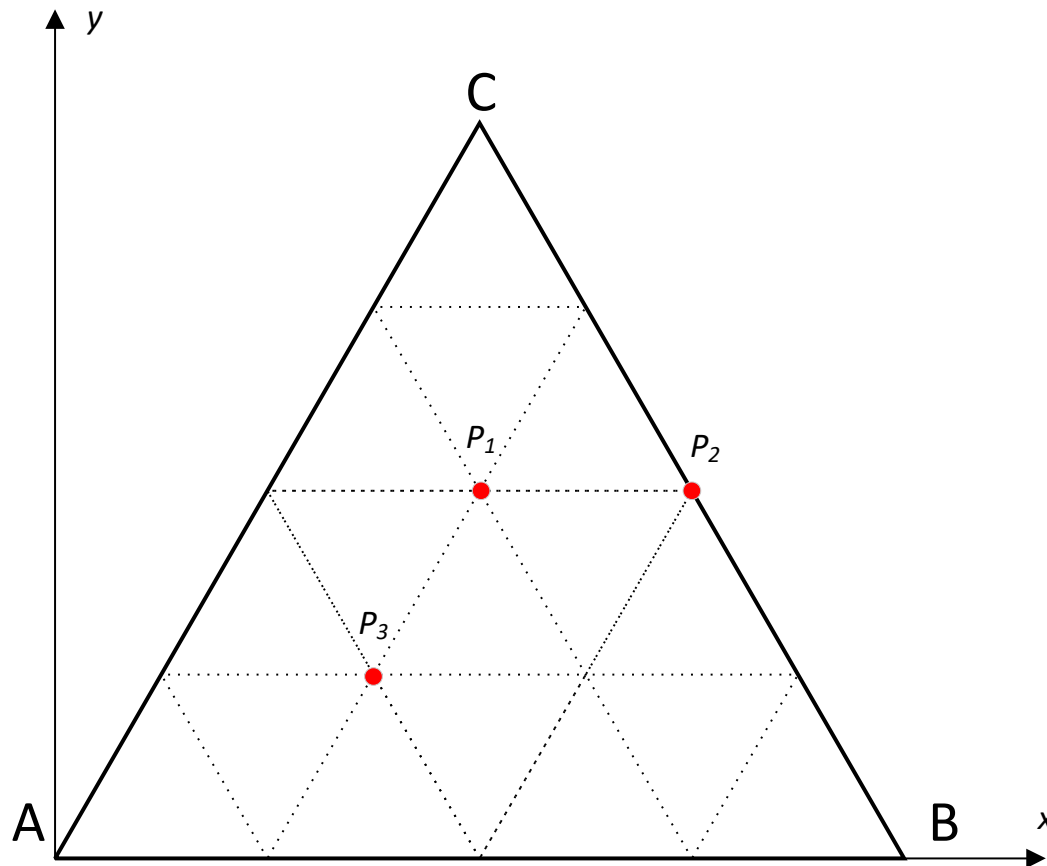


$$\begin{aligned} A &= (0, 0) \\ B &= (1.0, 0) \\ C &= (0.5, 0.866) \\ P_1 &= (0.5, 0.433) \end{aligned}$$

$$\begin{aligned} \alpha &= ? \\ \beta &= ? \\ \gamma &= ? \end{aligned}$$



Coordinate Example I

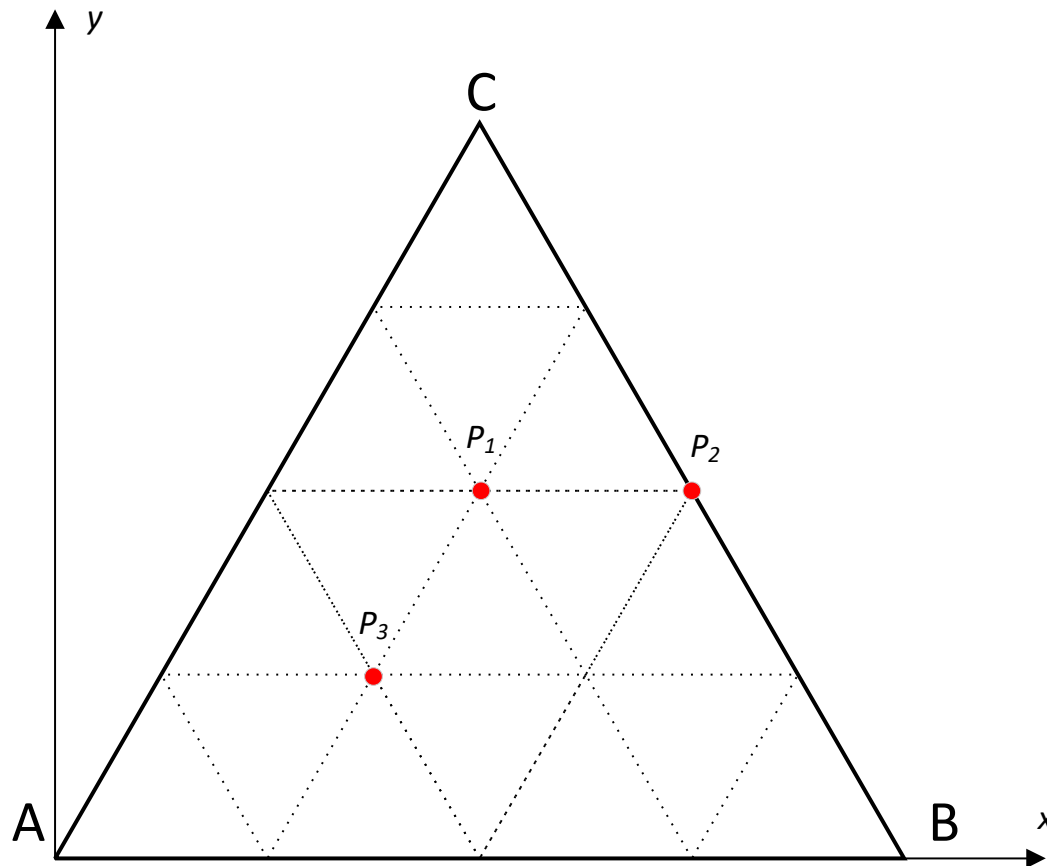


$$\begin{aligned}A &= (0, 0) \\B &= (1.0, 0) \\C &= (0.5, 0.866) \\P_1 &= (0.5, 0.433)\end{aligned}$$

$$\begin{aligned}\alpha &= 0.25 \\ \beta &= 0.25 \\ \gamma &= 0.5\end{aligned}$$



Coordinate Example II



$$A = (0, 0)$$

$$B = (1.0, 0)$$

$$C = (0.5, 0.866)$$

$$P_2 = (0.75, 0.433)$$

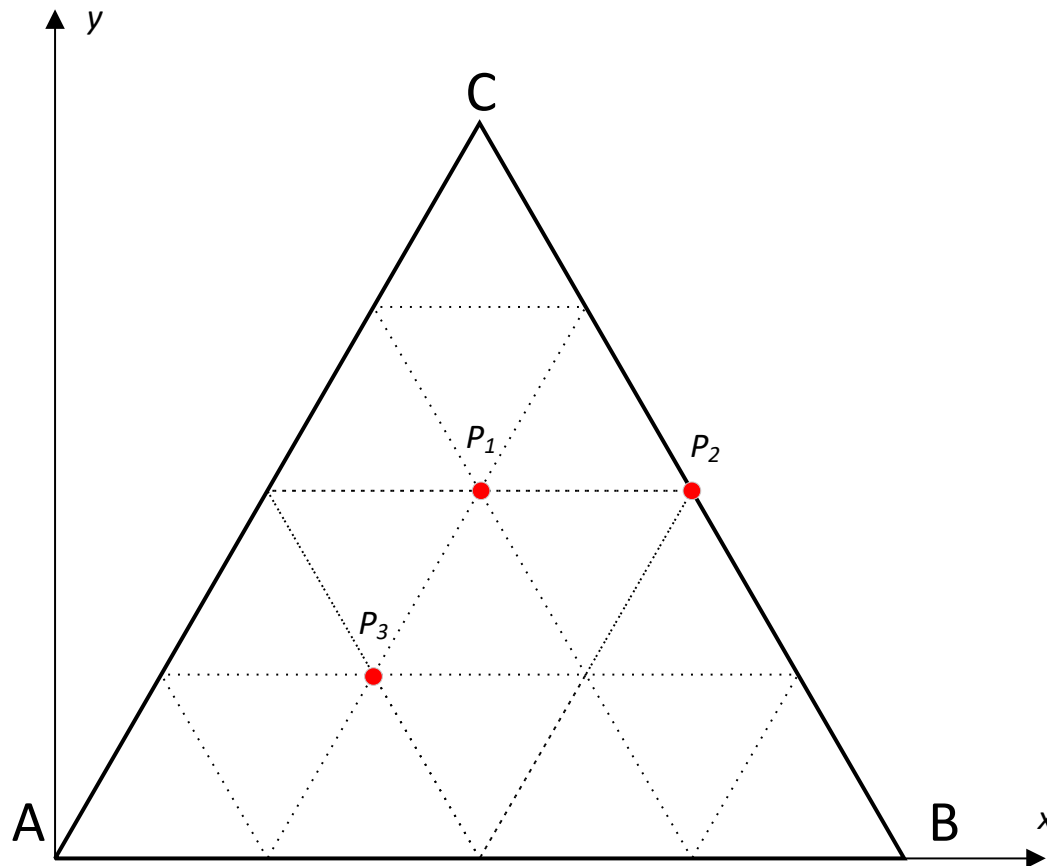
$$\alpha = ?$$

$$\beta = ?$$

$$\gamma = ?$$



Coordinate Example II



$$A = (0, 0)$$

$$B = (1.0, 0)$$

$$C = (0.5, 0.866)$$

$$P_2 = (0.75, 0.433)$$

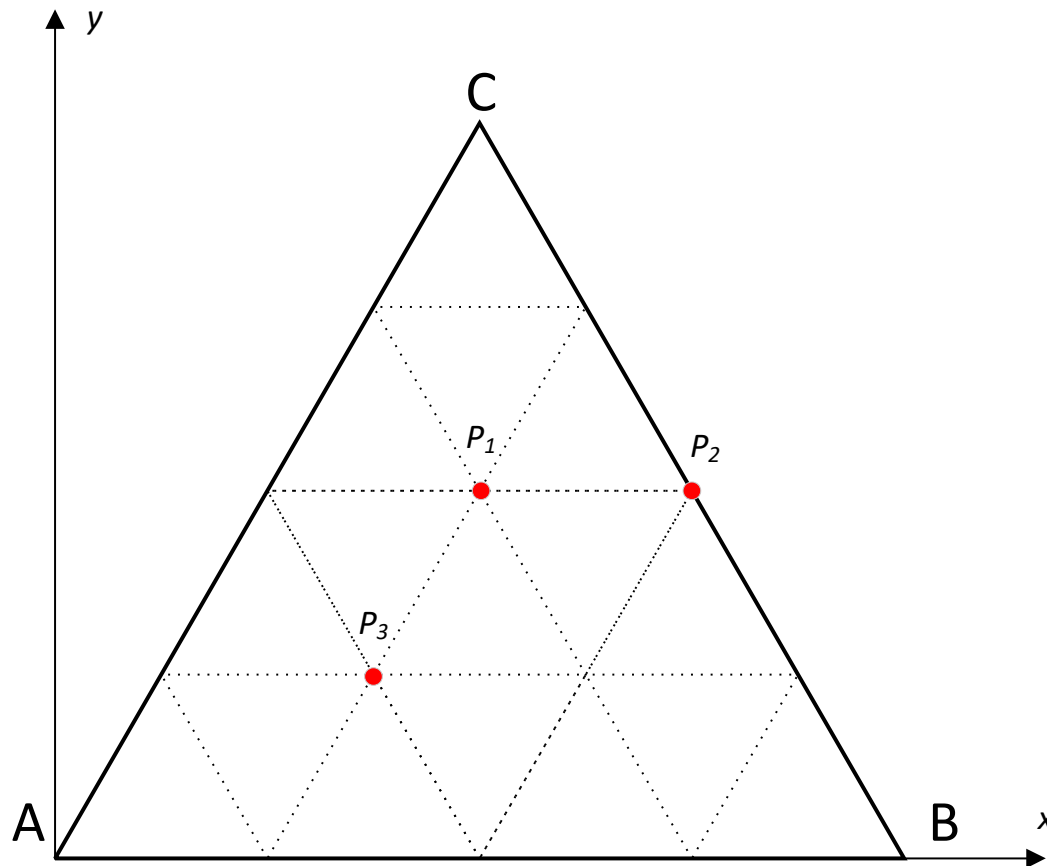
$$\alpha = 0$$

$$\beta = 0.5$$

$$\gamma = 0.5$$



Coordinate Example III

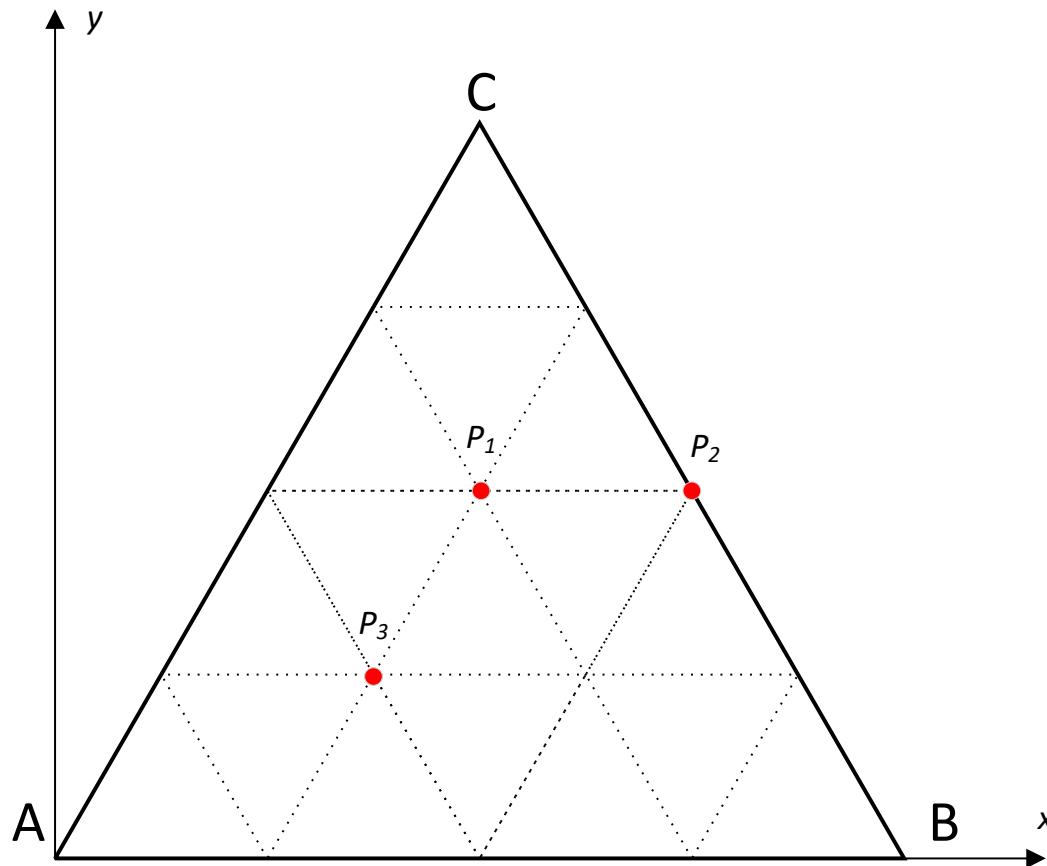


$$\begin{aligned} A &= (0, 0) \\ B &= (1.0, 0) \\ C &= (0.5, 0.866) \\ P_3 &= (0.375, 0.216) \end{aligned}$$

$$\begin{aligned} \alpha &= ? \\ \beta &= ? \\ \gamma &= ? \end{aligned}$$



Coordinate Example III



$$\begin{aligned} A &= (0, 0) \\ B &= (1.0, 0) \\ C &= (0.5, 0.866) \\ P_3 &= (0.375, 0.216) \end{aligned}$$

$$\begin{aligned} \alpha &= 0.5 \\ \beta &= 0.25 \\ \gamma &= 0.25 \end{aligned}$$



3D Triangles

- Barycentric coordinates can be used in 3D.

$$\mathbf{p} = (1 - \beta - \gamma)\mathbf{a} + \beta\mathbf{b} + \gamma\mathbf{c}.$$

$$\mathbf{n} = (\mathbf{b} - \mathbf{a}) \times (\mathbf{c} - \mathbf{a}).$$

$$\text{area} = \frac{1}{2} \|(\mathbf{b} - \mathbf{a}) \times (\mathbf{c} - \mathbf{a})\|.$$

$$\alpha = \frac{\mathbf{n} \cdot \mathbf{n}_a}{\|\mathbf{n}\|^2},$$

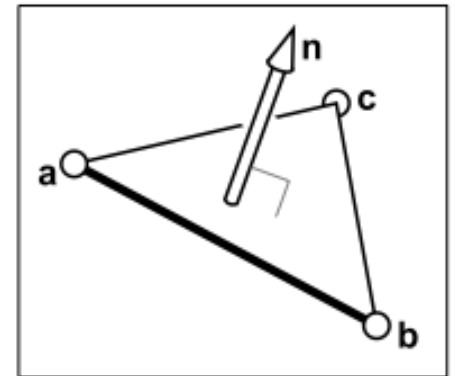
$$\beta = \frac{\mathbf{n} \cdot \mathbf{n}_b}{\|\mathbf{n}\|^2},$$

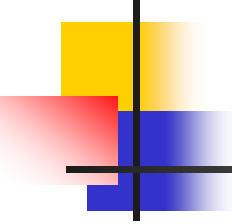
$$\gamma = \frac{\mathbf{n} \cdot \mathbf{n}_c}{\|\mathbf{n}\|^2},$$

$$\mathbf{n}_a = (\mathbf{c} - \mathbf{b}) \times (\mathbf{p} - \mathbf{b}),$$

$$\mathbf{n}_b = (\mathbf{a} - \mathbf{c}) \times (\mathbf{p} - \mathbf{c}),$$

$$\mathbf{n}_c = (\mathbf{b} - \mathbf{a}) \times (\mathbf{p} - \mathbf{a}).$$





Questions



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